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DataInterpolations.jl: Fast Interpolations of 1D data

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Summary

Interpolations are used to estimate values between known data points using an approximate continuous function.DataInterpolations.jl is a Julia (Bezanson et al., 2017) package containing 1D implementations of some of the most commonly used interpolation functions. These include Constant Interpolation, Linear Interpolation, Quadratic Interpolation, Lagrange Interpolation (**?**), Quadratic Splines, Cubic Splines (Schoenberg, 1988), Akima Splines (Akima, 1970), B-Splines (Curry & Schoenberg, 1988) (de Boor, 1972) and Regression based B-Splines. Along with these, the package also has methods to fit parameterized curves with the data points and Tikhonov regularization (Tikhonov, 1943) (Petters, 2021) for obtaining smooth curves. The package also provides functionality to compute integrals and derivatives upto second order for those interpolations methods.

Statement of need

Interpolations are a very important component of many modeling workflows. In many models, inputs which are sampled or measured need to be represented as a continuous function or a smooth curve for simulation. In many scientific machine learning workflows, we need interpolations of data to learn continuous models. There already have been a few interpolation packages in Julia like Interpolations.jl but it has a limitation of assuming uniformly spaced data which is not usually the case with data collected from real world. DataInterpolations.jl provides fast interpolation methods for arbitrary spaced 1D data with a consistent and simple interface. It is also automatic differentiation friendly. It can also be used symbolically with Symbolics.jl (**?**) and plugged into models defined using ModelingToolkit.jl (Ma et al., 2021).

Example

The following tutorials in the documentation 1 provides how to define each of the interpolation methods and compute the value at any point. 2 provides explanation for using the interface and interpolated objects for evaluating at any point, computing the derivative at any point and computing the integral between any two points.

A simple demonstration here:

```julia
using DataInterpolations

# Dependent variable
u = [14.7, 11.51, 10.41, 14.95, 12.24, 11.22]

# Independent variable
t = [0.0, 62.25, 109.66, 162.66, 205.8, 252.3]
```

```
A1 = CubicSpline(u, t)

# For interpolation do, A(t)
A1(100.0)

# derivative
## first order
DataInterpolations.derivative(A1, 1.0, 1)

## second order
DataInterpolations.derivative(A1, 1.0, 2)

# integral
DataInterpolations.integral(A1, 1.0, 5.0)
```

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